Learning With Errors (LWE)

The math under ML-KEM, the post-quantum half of ferrokey's lock

1. The idea in one picture

I pick a secret list of numbers. I give you lots of clues: "this mix of your secret numbers adds up to *about* this much." If the clues were exact, you'd solve them like any system of equations from 21-241. Because every clue is off by a tiny random amount (the **noise**, or **error**), solving them becomes hopelessly hard, even for a quantum computer.

That's the whole thing: **linear equations + a little noise = a hard problem**.

2. Without noise it's easy

Work mod $q = 17$ (all arithmetic wraps around at 17). Secret $s = (3, 5)$. Four clue rows:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 1 \\ 2 & 7 \\ 5 & 3 \end{pmatrix}, \qquad As = \begin{pmatrix} 13 \\ 0 \\ 7 \\ 13 \end{pmatrix} \pmod{17}.$$

Given A and As, take the first two rows: $x + 2y = 13$, $4x + y = 0$. The second gives $y = -4x$; substituting, $-7x \equiv 13$, so $7x \equiv 4$. Since $7 \cdot 5 = 35 \equiv 1$, the inverse of 7 is 5, and $x = 20 \equiv 3$, $y = -12 \equiv 5$. Secret recovered. This is just Gaussian elimination mod 17. (The Hill cipher, $c = Km \bmod 26$, falls to the same attack.)

3. Add noise and elimination falls apart

Now publish $b = As + e$ with small noise $e = (1, -1, 0, 1)$:

$$b = \begin{pmatrix} 14 \\ 16 \\ 7 \\ 14 \end{pmatrix}.$$

Same steps on rows 1–2: $x + 2y = 14$, $4x + y = 16$, so $y = 16 - 4x$, $-7x \equiv -18$, $7x \equiv 1$, $x \equiv 5$, $y \equiv 13$. Guess: $(5, 13)$. Truth: $(3, 5)$. Completely wrong.

Check it on row 3: $2 \cdot 5 + 7 \cdot 13 = 101 \equiv 16$, but $b_3 = 7$. Off by 9, not by a small amount.

Why: elimination multiplies and adds rows. Each step multiplies the noise too, and mod q a few multiplications turn "off by 1" into "off by anything." The exact-algebra tools you know stop working.

With 2 unknowns you could still just try all $17^2 = 289$ guesses. Real sizes make that impossible (Section 6).

4. The definition

Pick a modulus q, a dimension n and a number of samples m.

- Secret $s \in \mathbb{Z}_q^n$.
- Public matrix $A \in \mathbb{Z}_q^{m \times n}$, uniformly random.
- Noise $e \in \mathbb{Z}^m$, every entry small (e.g. in $\{-2, \ldots, 2\}$).
- Publish $(A,\ b = As + e \bmod q)$.

Search-LWE: given (A, b), find s. **Decision-LWE:** tell (A, b) apart from (A, random). Both are believed hard. Regev (2005) proved that breaking LWE on average is at least as hard as solving certain lattice problems in the *worst* case, which is why cryptographers trust it.

5. Locking one bit with LWE

Your public key is (A, b) from Section 3. **Your private key** is s.

To send bit μ: pick a random 0/1 vector r (which clue rows to add up), then send

$$u = A^{\mathsf{T}} r, \qquad v = b^{\mathsf{T}} r + \mu \cdot \lfloor q/2 \rfloor.$$

To open: compute $v - s^{\mathsf{T}} u$. Because $b = As + e$,

$$v - s^{\mathsf{T}} u = \underbrace{r^{\mathsf{T}} e}_{\text{small}} + \mu \lfloor q/2 \rfloor.$$

Near 0 means $\mu = 0$; near $q/2$ means $\mu = 1$.

Worked: $q = 17$, $\lfloor q/2 \rfloor = 8$, $r = (1, 0, 1, 1)$ (add rows 1, 3, 4).

- $u = (1{+}2{+}5,\ 2{+}7{+}3) = (8, 12)$.
- $b^{\mathsf{T}} r = 14 + 7 + 14 = 35 \equiv 1$. Send $\mu = 1$: $v = 1 + 8 = 9$.
- Open: $s^{\mathsf{T}} u = 3 \cdot 8 + 5 \cdot 12 = 84 \equiv 16$. Then $v - 16 = -7 \equiv 10$.
- 10 is 2 away from 8 and 7 away from 0 (or 17), so the bit is **1**. ✓

The leftover 10 is $8 + 2$, and the 2 is exactly the noise in the rows we added: $e_1 + e_3 + e_4 = 1 + 0 + 1$.

You can subtract $s^{\mathsf{T}} u$ exactly because you know s, so only a little noise is left. An attacker without s sees u and v as random-looking numbers. Opening fails only if the summed noise passes $q/4$; real parameters make that chance about 2^{-164}.

6. From this toy to ML-KEM-768 (what ferrokey uses)

- **Bigger numbers:** $q = 3329$, noise in $\{-2, \ldots, 2\}$.
- **Module-LWE:** each entry of A is a polynomial with 256 coefficients instead of a single number, and A is 3×3 of those. Effectively 768 unknowns, so brute force means about 3329^{768} guesses. Polynomials make keys smaller and arithmetic fast.
- **Key, not message:** instead of one bit, it locks a 256-bit random key; AES-256-GCM then encrypts the actual message.
- **Public key size:** 1184 bytes. Add X25519's 32 bytes and you get the 1216-byte X-Wing key in `identity.ts`.

7. Why a quantum computer doesn't help

Shor's algorithm breaks RSA and elliptic curves by finding a **period**, a pattern that repeats ($a^x \bmod n$ cycles). LWE has no such hidden cycle to find. The best known attacks, quantum or not, are lattice

searches whose cost grows exponentially with the dimension. Quantum computers give at most a modest speedup there, already priced into the parameters.

Check yourself (answer in chat)

1. In one sentence: why can't an attacker just run Gaussian elimination on (A, b)?
2. In Section 5, suppose the noise could be as big as ± 3 per row. What could go wrong when opening, and why?
3. Why does the person holding s only see a *little* noise, while the attacker's elimination blows it up?